

Bounded recovery for modular spectral averages

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Abstract

We prove a bounded spectral recovery theorem for a von Neumann algebra with a faithful normal state whose centralizer consists only of scalars. A positive averaged squared norm for an arbitrary bounded Hilbert-space operator on shrinking modular spectral bands can be recovered on uniformly bounded algebra elements with shrinking spectral support. We apply this theorem to spectral intertwining rigidity and obtain an alternative proof of bicentralizer triviality for type III₁ factors with separable predual.

1 Introduction

Spectral localization in the standard Hilbert space of a von Neumann algebra does not by itself produce uniformly bounded elements of the algebra. This distinction matters when an argument starts with spectral vectors but its asymptotic identities apply only to bounded operator sequences. Our main analytical result, [Theorem 4.1](#), provides a bounded recovery principle for this situation.

Let M be a von Neumann algebra with a faithful normal state ϕ . Write M_ϕ for the fixed-point algebra of its modular automorphism group σ^ϕ . We call ϕ ergodic when $M_\phi = \mathbb{C}1$. On the standard Hilbert space $H = L^2(M)$, put $\xi = \phi^{1/2}$, $D = \log \Delta_\phi$, and $U_t = e^{itD}$. For an arbitrary fixed $T \in \mathcal{B}(H)$, consider unit vectors supported in spectral intervals of D shrinking to a real number s . [Theorem 4.1](#) shows that a positive limiting averaged value of $\|TU_t h\|^2$ on these vectors can be recovered as a positive value of $\|Tv\xi\|^2$ on a sequence of uniformly bounded $v \in M$. The recovered vectors remain in intervals shrinking to s , with widths at most four times the original widths. The averaging uses a fixed translation-invariant mean obtained from symmetric interval averages. The exact quantifiers and mean are given in [Theorems 3.4](#) and [4.1](#).

The only structural assumption in this recovery theorem is $M_\phi = \mathbb{C}1$; neither a bicentralizer condition nor a second action is required. This assumption supplies a broad class of applications: Marrakchi and Vaes proved that ergodic faithful normal states form a dense G_δ in the normal state space of every type III₁ factor with separable predual [[14](#), Theorem A].

Bicentralizers and prior work. We use the predual convention

$$(w\phi)(y) = \phi(yw), \quad (\phi w)(y) = \phi(wy).$$

The bicentralizer $\text{BC}(M, \phi)$ consists of those $a \in M$ for which $[a, w_n] \rightarrow 0$ strongly whenever (w_n) is uniformly bounded and $\|w_n\phi - \phi w_n\|_{M_*} \rightarrow 0$. Connes' bicentralizer conjecture asserts that this algebra is scalar for every type III₁ factor with separable predual. Connes developed the problem in his classification program for injective factors [[3](#), [4](#)]. Haagerup proved the amenable case, completing the uniqueness theorem for the injective type III₁ factor [[5](#)].

Haagerup also related triviality to the existence of a faithful normal state with irreducible centralizer, meaning $(M_\phi)' \cap M = \mathbb{C}1$ [5, Theorem 3.1]. Together with Popa's work on maximal abelian subalgebras [15], this connects the problem to expected subalgebras; the expected-masa equivalence is stated in [8, Corollary 3.6]. Important nonamenable cases include Houdayer's theorem for free Araki–Woods factors [7] and the semisolid factors treated by Houdayer and Isono [8, Theorem 3.7]. An irreducible centralizer in these results is different from the scalar centralizer assumed in our recovery theorem.

Houdayer and Isono proved that a nonscalar bicentralizer of a separable type III_1 factor is itself a type III_1 factor and equals its own bicentralizer for the restricted state [8, Proposition 3.4 and Theorem 3.5]. Ando, Haagerup, Houdayer, and Marrakchi established the canonical relative bicentralizer flow [1, Theorem A], following the flow construction of Haagerup and its independent discovery by Marrakchi. Marrakchi proved that every non-identity time of the absolute flow is ergodic, without a fullness assumption [11, Theorem D]. His subsequent work develops the binormal-state and relative-core machinery [12, 13].

The absolute application and the recent announcement. Houdayer and Marrakchi have announced triviality of the bicentralizer for every type III_1 factor admitting a faithful normal state, with no separability condition in the theorem statement [9, Theorem G]. The application proved here is an alternative proof of the separable-predual instance of that conclusion.

Theorem 1.1. *Let M be a type III_1 factor with separable predual and let ϕ be any faithful normal state on M . Then*

$$\text{BC}(M, \phi) = \mathbb{C}1.$$

By [8, Corollary 3.6], Theorem 1.1 also implies that every type III_1 factor with separable predual admits a maximal abelian subalgebra with a faithful normal conditional expectation. This is an existence statement for the factor; the expectation need not preserve the scalar-centralizer state used in bounded recovery.

The independent analytical contribution is the bounded recovery theorem, which applies to arbitrary T without assuming a bicentralizer intertwining law. Houdayer and Marrakchi obtain the same symmetric spectral relation used below through multiplication isometries and a singular KMS state on the continuous core [9, Section 8.4, Propositions 8.8–8.10]. Here modular averaging and bounded recovery lead to that relation through two interval estimates.

Houdayer and Marrakchi explicitly disclose assistance from *GPT-5.6 Sol* and *GPT-6 Astra* [9, p. 7]. They credit Sol with counterexample exploration and an initial strict-outerness argument, and Astra with the singular tracial-state ingredient in their resonance strategy. They describe checking and rewriting the proofs and retaining responsibility for them. These are their stated contributions to that work, whose relative tracial-action theory is also developed in the paper.

The rigidity statement. A continuous action on M means a u -continuous action on its automorphism group. Bounded recovery gives the following spectral proof of rigidity.

Theorem 1.2 (Spectral intertwining rigidity). *Let M be a von Neumann algebra with a faithful normal state ϕ such that $M_\phi = \mathbb{C}1$. Let $b : \mathbb{R} \curvearrowright M$ be a continuous ϕ -preserving action. Suppose that, for every $s \in \mathbb{R}$, every uniformly bounded sequence (w_n) in M , and every $a \in M$,*

$$\|\phi w_n - e^s w_n \phi\|_{M_*} \longrightarrow 0 \implies w_n a - b_s(a) w_n \longrightarrow 0 \text{ strongly.} \quad (1.1)$$

Then $b_s = \text{id}_M$ for every $s \in \mathbb{R}$.

The conclusion also follows from the announced absolute theorem. Indeed, at $s = 0$, hypothesis (1.1) implies $\text{BC}(M, \phi) = M$. If $M \neq \mathbb{C}1$, its scalar centralizer makes it a type III_1 factor by [14, Lemma 2.1]; the announced theorem would then force $M = \mathbb{C}1$. Our proof instead derives Theorem 1.2 directly from bounded recovery and uses it to prove Theorem 1.1. For that application to a self-bicentralizing factor, the modular centralizer is scalar, the bicentralizer flow satisfies (1.1), and that flow is ergodic. A trivial ergodic action has scalar underlying algebra.

Recovery and the spectral argument. The recovery proof combines modular averaging with the fourth-moment and polar-cutoff devices of Haagerup and Musat [6, Equation (2.10), Lemma 2.6, and Remark 2.8]. The strategy of recovering bounded witnesses by randomization, fourth-moment control, and truncation appears in the tracial spectral-gap argument of Ioana and Vaes [10, pp. 127–130]. Related fourth-moment control underlies Marrakchi’s binormal-state implementation [12, Section 2, Claim 2.2]. Here modular orbit averages supply the moment bounds, and a final Fourier filter restores the shrinking spectral bands. The averaging time is chosen separately for each initial bounded representative of a spectral vector; these representatives need not be uniformly bounded. Norm convergence in the predual controls a varying second-moment operator, and independent circle phases give a uniform fourth-moment bound. One fixed polar truncation threshold then suffices. Fourier localization restores the spectral support without increasing operator norms by a factor depending on the width of the spectral interval. The Fourier operations belong to the standard spectral-subspace framework for automorphism groups [2]; the bounds needed here are proved explicitly.

For the spectral application, implement the state-preserving action by $V_s(a\xi) = b_s(a)\xi$ and write $V_s = e^{isQ}$. These unitaries commute with U_t , so (D, Q) has a joint spectral support $\mathcal{S} \subset \mathbb{R}^2$. The recovered bounded sequences let (1.1) force averaged intertwining defects to vanish on bands shrinking to a fixed modular energy. A spectral partition then gives an interval estimate without requiring a rate uniform in that energy. Applying this estimate twice, with the operators interchanged and the time reversed, gives

$$\exp(i(r\ell + qp)) = 1 \quad ((r, p), (q, \ell) \in \mathcal{S}).$$

This symmetric relation is the final rigidity mechanism. It forces the joint support to be countable or to lie on a coordinate axis. Scalarity of the modular centralizer excludes nonzero modular eigenvalues, by the bounded-eigenoperator extraction and KMS argument of [14, Lemma 2.1], leaving only the axis on which the flow generator vanishes.

Organization. Section 2 records the established bicentralizer inputs and reduces Theorem 1.1 to Theorem 1.2. Section 3 proves the modular and averaging preliminaries. Section 4 proves Theorem 4.1, and Section 5 completes the spectral argument.

2 Bicentralizers and the reduction

We recall the definition to fix both the topology and the quantifiers. For a faithful normal state ϕ , put $\|x\|_\phi = \phi(x^*x)^{1/2}$. The bicentralizer consists of those $a \in M$ for which, for every $\varepsilon > 0$, there is $\delta > 0$ such that

$$u \in \mathcal{U}(M), \quad \|u\phi - \phi u\|_{M_*} < \delta \quad \implies \quad \|u^*au - a\|_\phi < \varepsilon.$$

This is Haagerup’s unitary formulation of the usual bicentralizer [5, proof of Lemma 1.2]. Equivalently, its elements asymptotically commute strongly with every uniformly bounded

sequence that asymptotically centralizes ϕ in predual norm. We use established structural results in this equivalent form.

Proposition 2.1 (Established bicentralizer results). *Let M be a type III₁ factor with separable predual and ϕ a faithful normal state.*

- (i) *If $B = \text{BC}(M, \phi)$ is nonscalar, then B is a type III₁ subfactor with a normal ϕ -preserving conditional expectation, and*

$$\text{BC}(B, \phi|_B) = B.$$

- (ii) *There is a continuous ϕ -preserving action $\beta^\phi : \mathbb{R}_+^* \curvearrowright \text{BC}(M, \phi)$ characterized by*

$$\|w_n \phi - \lambda \phi w_n\|_{M_*} \longrightarrow 0 \implies w_n a - \beta_\lambda^\phi(a) w_n \longrightarrow 0 \quad * \text{-strongly}$$

for every $\lambda > 0$, every uniformly bounded sequence (w_n) in M , and every $a \in \text{BC}(M, \phi)$.

- (iii) *The fixed-point algebra of β^ϕ is $\mathbb{C}1$.*

Sources and hypotheses. The restricted-state identity in (i) is [8, Proposition 3.4]; the type dichotomy and expectation are [8, Theorem 3.5]. The bicentralizer is globally invariant under the modular group by [8, Proposition 3.3], so its canonical expectation preserves ϕ . These results impose no amenability or fullness hypothesis.

For (ii), apply [1, Theorem A(ii)] to the identity inclusion $M \subset M$. Continuity is part of the flow construction. The expectation-preservation identity in [1, Theorem A(iv)] specializes to preservation of ϕ , because the relative commutant is $\mathbb{C}1$.

Finally, [11, Theorem D] states, for any σ -finite type III₁ factor, that every β_λ^ϕ with $\lambda \neq 1$ is ergodic. This is stronger than (iii). Fullness is not assumed in this Theorem. $\square$

Proof of Theorem 1.1, assuming Theorem 1.2. Suppose that $\text{BC}(M, \phi)$ is nonscalar. Replace (M, ϕ) by $(B, \phi|_B)$ using Theorem 2.1(i). The restricted state is faithful and normal. The predual is still separable: if $E : M \rightarrow B$ is the normal expectation, $B_* \ni \rho \mapsto \rho \circ E \in M_*$ is an isometric embedding. We may therefore assume

$$\text{BC}(M, \phi) = M, \quad M \neq \mathbb{C}1.$$

Every unitary $u \in M_\phi$ centralizes ϕ exactly. The unitary definition of the bicentralizer gives $u^* a u = a$ for every $a \in M$. Since unitaries span M_ϕ , and M is a factor, it follows that $M_\phi = \mathbb{C}1$.

Set $b_s = \beta_{e^{-s}}^\phi$. The premise of (1.1) is equivalent to

$$\|w_n \phi - e^{-s} \phi w_n\|_{M_*} \longrightarrow 0.$$

Theorem 2.1(ii) therefore gives exactly (1.1), and also continuity and state preservation. By Theorem 1.2, b is trivial. By Theorem 2.1(iii), it is ergodic. Hence $M = \mathbb{C}1$, contradicting the reduction. The contradiction applies to every original pair (M, ϕ) . $\square$

3 Modular localization and averaging

The recovery argument requires two kinds of control: modular spectral localization must imply a predual norm estimate, and modular averaging must control the moments of an operator even when its norm is large. We establish these facts here. The two norm limits in Theorem 3.3 will control different terms in the fourth-moment expansion in Section 4.

Throughout this Section M is a von Neumann algebra with a faithful normal state ϕ . Let $H = L^2(M)$ be its standard Hilbert space, $\xi = \phi^{1/2}$ its cyclic separating unit vector, and

$$L(a)h = ah, \quad R(a)h = ha, \quad D = \log \Delta_\phi, \quad U_t = e^{itD}.$$

Write E_D for the spectral resolution of D and J for the modular conjugation. Inner products are linear in the first variable. We use the standard modular theory of [16]. We write $\sigma_t = \sigma_t^\phi$. Standard modular theory gives

$$U_t(a\xi) = \sigma_t(a)\xi, \quad e^{D/2}a\xi = \xi a, \quad R(a) = JL(a)^*J. \quad (3.1)$$

Indeed, $M\xi$ is contained in the domain of the closed Tomita operator $S = J\Delta_\phi^{1/2}$, with $S(a\xi) = a^*\xi$. Thus $\Delta_\phi^{1/2}a\xi = Ja^*\xi = \xi a$. These standard-form conventions are also those of [1, Section 2.1].

Lemma 3.1 (From modular localization to predual norm). *For $w \in M$ and $s \in \mathbb{R}$,*

$$\|\phi w - e^s w \phi\|_{M_*} \leq (1 + e^{s/2})\|(e^{D/2} - e^{s/2})w\xi\|. \quad (3.2)$$

Consequently, if (w_n) is uniformly bounded and the D -spectral support of $w_n\xi$ is contained in $[s - d_n, s + d_n]$ with $d_n \rightarrow 0$, then the left side of (3.2) tends to zero.

Proof. Put $c = e^{s/2}$ and $\eta = \xi w - cw\xi$. By (3.1), $\eta = (e^{D/2} - c)w\xi$, and $J\eta = w^*\xi - c\xi w^*$. For $\|y\| \leq 1$, commutation of the left and right actions gives

$$\langle y\xi, \xi w^* \rangle = \langle y(\xi w), \xi \rangle.$$

Starting from $\phi(wy) = \langle y\xi, w^*\xi \rangle$, replace $w^*\xi$ by $c\xi w^*$, and then ξw by $cw\xi$. The total error is at most $(1 + c)\|\eta\|$, while the resulting expression is $c^2\phi(yw)$. Taking the supremum over y proves the estimate. The last assertion follows from

$$\|(e^{D/2} - e^{s/2})w_n\xi\| \leq \sup_{|u-s| \leq d_n} |e^{u/2} - e^{s/2}| \|w_n\xi\|. \quad \square$$

Lemma 3.2 (Smooth spectral localization). *Let $f \in C_c^\infty(\mathbb{R})$, with Fourier convention $f(u) = \int_{\mathbb{R}} g(t)e^{itu} dt$. For $y \in M$, the weak integral*

$$y_f = \int_{\mathbb{R}} g(t)\sigma_t(y) dt$$

belongs to M and satisfies $\|y_f\| \leq \|g\|_1\|y\|$ and $y_f\xi = f(D)y\xi$. If $a \in M$ and $a\xi$ has compact D -spectral support, there is $a^\flat \in M$ such that

$$a\xi = \xi a^\flat. \quad (3.3)$$

The same Fourier construction applies to any continuous state-preserving action and its unitary implementation.

Proof. The inverse Fourier transform g is integrable. The bounded weak integral is defined by integration against normal functionals, and its operator norm has the stated bound. Applying it to ξ and using the spectral theorem gives $y_f\xi = f(D)y\xi$. This uses no operator norm continuity of the orbit.

If K is a compact support for $a\xi$, choose $f \in C_c^\infty(\mathbb{R})$ equal to $e^{-u/2}$ on a neighborhood of K , and put $a^\flat = a_f$. Then (3.1) and functional calculus give

$$\xi a^\flat = e^{D/2}a^\flat\xi = e^{D/2}f(D)a\xi = a\xi.$$

All vectors in this computation belong to the domain of $e^{D/2}$; in particular the filtered vector has compact spectral support. The Fourier argument for another state-preserving action is identical. $\square$

For the rest of this Section assume $M_\phi = \mathbb{C}1$. Let μ_k be uniform probability measure on $[-k, k]$, for $k \in \mathbb{N}$.

Lemma 3.3 (Two norm ergodic limits). *For every $y \in M$ and $\psi \in M_*$,*

$$\int \sigma_t(y) \xi \, d\mu_k(t) \longrightarrow \phi(y) \xi \quad \text{in } H, \quad (3.4)$$

$$\int \psi \circ \sigma_{-t} \, d\mu_k(t) \longrightarrow \psi(1) \phi \quad \text{in } M_*. \quad (3.5)$$

Proof. The unitary mean ergodic theorem gives a norm limit for the vectors in (3.4). The corresponding operator averages have norm at most $\|y\|$. For each fixed u , translating the interval changes their averages in operator norm by at most $|u|\|y\|/k$ for all sufficiently large k . Every weak operator cluster point is consequently fixed by σ , hence scalar. Evaluation under ϕ identifies it as $\phi(y)1$. This identifies the Hilbert norm limit and proves (3.4).

The action on M_* is norm continuous on each functional. For positive normal functionals this follows from their realization as vector functionals in standard form and the strong continuity of U ; linear decomposition gives the general case. The predual averages are therefore Bochner integrals and are contractions. Let

$$\mathcal{N} = \overline{\text{span}}^{\|\cdot\|} \{\rho \circ \sigma_u - \rho : \rho \in M_*, u \in \mathbb{R}\}.$$

Its annihilator in $(M_*)^* = M$ is $M_\phi = \mathbb{C}1$. The Hahn–Banach theorem implies $\mathcal{N} = \{\psi \in M_* : \psi(1) = 0\}$. The averages tend to zero in norm on each generator of $\mathcal{N}$, by the interval boundary estimate, and hence on all of $\mathcal{N}$ by contractivity. They fix ϕ . Apply this to $\psi - \psi(1)\phi$ to obtain (3.5). $\square$

Fix a free ultrafilter ω on $\mathbb{N}$. For a bounded continuous scalar function f define

$$m_t f(t) = \lim_{k \rightarrow \omega} \int f(t) \, d\mu_k(t).$$

This positive normalized mean is invariant under translations and under $t \mapsto -t$. For a bounded continuous H -valued function F , put

$$\|F\|_{\text{av}} = (m_t \|F(t)\|^2)^{1/2}.$$

The triangle inequality follows from the corresponding inequalities for the finite interval integrals. Thus this is a seminorm.

Lemma 3.4 (Operator averaging). *For $S \in \mathcal{B}(H)$, define $P(S)$ by*

$$\langle P(S)h, k \rangle = m_t \langle U_t^* S U_t h, k \rangle.$$

Then P is a positive unital contraction, $P(S)$ commutes with every spectral projection of D , and

$$P(L(a)) = \phi(a)1 \quad (a \in M). \quad (3.6)$$

In particular,

$$\|t \mapsto S U_t h\|_{\text{av}}^2 = \langle P(S^* S)h, h \rangle. \quad (3.7)$$

Proof. The bounded sesquilinear form defines $P(S)$, with $\|P(S)\| \leq \|S\|$. Positivity and unitality follow directly from the mean. Translation invariance gives $U_u^* P(S) U_u = P(S)$ for every u , hence commutation with the spectral projections of D . For $S = L(a)$ the averages and their weak limits belong to the weakly closed algebra $L(M)$, since $U_t^* L(a) U_t = L(\sigma_{-t}(a))$. The fixed-point algebra is scalar, and evaluation at ξ gives (3.6). Equation (3.7) is the definition applied to $S^* S$. No normality of P is asserted or needed. $\square$

4 Recovering bounded operators from spectral vectors

[Theorem 4.1](#) converts a positive averaged quantity on narrow modular spectral bands into a positive quantity on uniformly bounded operators. Its proof uses both norm limits in [Theorem 3.3](#): the Hilbert-space limit controls one fourth-moment term, and the predual limit controls the other. No uniform operator norm bound is required for the initial representatives of the spectral vectors. Randomization, fourth-moment control, truncation, and selection of a bounded witness occur together in the tracial spectral-gap argument of Ioana and Vaes [10, proof of the appendix theorem, pp. 129–130]. Related fourth-moment control gives equi-integrability in Marrakchi’s binormal-state implementation [12, Section 2, Claim 2.2]. We use the fourth-moment and polar-cutoff devices of Haagerup and Musat [6, Lemma 2.6 and Remark 2.8], with moment bounds supplied by modular orbit averages against the given state. The final Fourier localization retains the shrinking spectral band.

Theorem 4.1 (Bounded recovery). *Let M be a von Neumann algebra with a faithful normal state ϕ such that $M_\phi = \mathbb{C}1$. Use the standard-space notation and the fixed mean m of [Theorem 3.4](#). Fix $T \in \mathcal{B}(H)$ and $s \in \mathbb{R}$. Suppose that $\delta_n > 0$ tends to zero and that $h_n \in H$ are unit vectors with D -spectral support in $[s - \delta_n, s + \delta_n]$. If*

$$\limsup_{n \rightarrow \infty} m_t \|TU_t h_n\|^2 > 0, \quad (4.1)$$

then there are a subsequence (n_j) , elements $v_j \in M$, and constants $C_ < \infty$ and $\eta > 0$ such that, for every j ,*

$$\|v_j\| \leq C_*, \quad \|Tv_j \xi\| \geq \eta,$$

and $v_j \xi$ has D -spectral support in $[s - 4\delta_{n_j}, s + 4\delta_{n_j}]$.

Proof. We use the same family of Fourier filters to localize the initial bounded approximations and to restore localization after polar truncation. Choose $f_0 \in C_c^\infty(\mathbb{R})$ such that

$$0 \leq f_0 \leq 1, \quad f_0 = 1 \text{ on } [-1, 1], \quad \text{supp } f_0 \subset [-2, 2],$$

and write $f_0(u) = \int_{\mathbb{R}} g_0(t) e^{itu} dt$, with g_0 a Schwartz function. For $\rho > 0$, set

$$f_\rho(u) = f_0((u - s)/\rho), \quad g_\rho(t) = \rho e^{-ist} g_0(\rho t).$$

Then $f_\rho(u) = \int_{\mathbb{R}} g_\rho(t) e^{itu} dt$ and

$$\|g_\rho\|_1 = \|g_0\|_1 =: C_1 \quad \text{for every } \rho > 0. \quad (4.2)$$

By [Theorem 3.2](#), the weak integral

$$\mathcal{F}_\rho(x) = \int_{\mathbb{R}} g_\rho(t) \sigma_t(x) dt \in M$$

satisfies

$$\mathcal{F}_\rho(x) \xi = f_\rho(D) x \xi, \quad \|\mathcal{F}_\rho(x)\| \leq C_1 \|x\|. \quad (4.3)$$

In particular, its vector has spectral support in $[s - 2\rho, s + 2\rho]$.

Approximation by bounded representatives. By [Equation \(4.1\)](#), after passing to a subsequence and reindexing, there is $\gamma > 0$ such that

$$m_t \|TU_t h_n\|^2 \geq 4\gamma \quad (n \geq 1).$$

Since $M\xi$ is dense in H , approximate h_n by $x_n \xi$ and apply $f_{\delta_n}(D)$. This multiplier fixes h_n and is a contraction. Thus the vectors $\mathcal{F}_{\delta_n}(x_n) \xi$ approximate h_n as closely as desired.

Normalize them, obtaining $y_n \in M$ with $\|y_n \xi\| = 1$ and spectral support in $[s - d_n, s + d_n]$, where $d_n = 2\delta_n$. Choose the approximations so that $\|y_n \xi - h_n\| \rightarrow 0$ and

$$m_t \|TU_t y_n \xi\|^2 \geq 2\gamma \quad (n \geq 1). \quad (4.4)$$

This is possible because the averaged seminorm in [Theorem 3.4](#) gives

$$|\|t \mapsto TU_t y_n \xi\|_{\text{av}} - \|t \mapsto TU_t h_n\|_{\text{av}}| \leq \|T\| \|y_n \xi - h_n\|.$$

Discard finitely many terms so that $d_n \leq 1$. The modular identity in [Equation \(3.1\)](#) gives

$$\phi(y_n^* y_n) = 1, \quad \phi(y_n y_n^*) = \|e^{D/2} y_n \xi\|^2 \leq e^{s+d_n}. \quad (4.5)$$

There is no asserted uniform bound on $\|y_n\|$.

Simultaneous defect and moment bounds. Fix n for the moment, and write $y = y_n$. With μ_k the uniform probability measure on $[-k, k]$, define

$$\begin{aligned} A_k &= \int \sigma_t(y^* y) d\mu_k(t), & B_k &= \int \sigma_t(y y^*) d\mu_k(t), \\ \psi(x) &= \phi(y^* x y), & \psi_k &= \int \psi \circ \sigma_{-t} d\mu_k(t). \end{aligned}$$

We seek simultaneous control of $\phi(A_k^2)$ and $\psi_k(B_k)$, which will bound the fourth moment after discretization. Here ψ is a normal state. State invariance gives the exact identity

$$\psi_k(x) = \int \phi(\sigma_t(y)^* x \sigma_t(y)) d\mu_k(t). \quad (4.6)$$

By the two ordinary norm limits in [Theorem 3.3](#),

$$A_k \xi \longrightarrow \xi, \quad \|\psi_k - \phi\|_{M_*} \longrightarrow 0.$$

Consequently $\phi(A_k^2) = \|A_k \xi\|^2 \rightarrow 1$. Although B_k depends on k , it satisfies

$$\|B_k\| \leq \|y\|^2, \quad \phi(B_k) = \phi(y y^*) \leq e^{s+1},$$

and hence

$$|\psi_k(B_k) - \phi(B_k)| \leq \|\psi_k - \phi\|_{M_*} \|y\|^2 \longrightarrow 0. \quad (4.7)$$

All these limits are taken with n fixed. For every sufficiently large k we may impose $\phi(A_k^2) \leq 2$ and $\|\psi_k - \phi\|_{M_*} \leq \|y\|^{-2}$ simultaneously. It follows that

$$\phi(A_k^2) + \psi_k(B_k) \leq 3 + e^{s+1}. \quad (4.8)$$

On the other hand, [Equation \(4.4\)](#) and the definition of m imply that the set of integers k for which

$$\int \|T \sigma_t(y) \xi\|^2 d\mu_k(t) > \frac{3}{2}\gamma$$

belongs to the fixed free ultrafilter. It therefore meets the cofinite set on which [Equation \(4.8\)](#) holds. Choose one such k , still for this fixed n .

We next replace this continuous average by a finite probability average. Put $z_t = \sigma_t(y)$. The three scalar functions

$$t \longmapsto \|T z_t \xi\|^2, \quad (t, u) \longmapsto \phi(z_t^* z_t z_u^* z_u), \quad (t, u) \longmapsto \phi(z_t^* z_u z_u^* z_t)$$

are continuous on the relevant compact interval or square. Indeed, $t \mapsto z_t$ is bounded and strong-star continuous, and products on bounded sets preserve this continuity. A common sufficiently fine partition of $[-k, k]$, using its normalized interval lengths as weights, therefore approximates all three integrals. The product weights approximate the double integrals defining $\phi(A_k^2)$ and $\psi_k(B_k)$. Thus there are finitely many times t_i and weights $p_i \geq 0$, with $\sum_i p_i = 1$, such that, writing

$$z_i = \sigma_{t_i}(y), \quad A = \sum_i p_i z_i^* z_i, \quad B = \sum_i p_i z_i z_i^*,$$

we have

$$\sum_i p_i \|T z_i \xi\|^2 \geq \gamma, \quad \phi(A^2) + \sum_i p_i \phi(z_i^* B z_i) \leq C, \quad C := 4 + e^{s+1}. \quad (4.9)$$

The finite averages may depend on n ; γ and C do not. In particular, no estimate on the rate in Equation (4.7) uniform in n has been used.

Randomization and truncation. For this finite average, let ϵ_i be independent random variables uniformly distributed on the complex unit circle, and set

$$z = \sum_i \sqrt{p_i} \epsilon_i z_i.$$

The vector $z\xi$ is supported in $[s - d_n, s + d_n]$. Independence and vanishing first moments give

$$\mathbb{E} \|T z \xi\|^2 = \sum_i p_i \|T z_i \xi\|^2 \geq \gamma.$$

The following expansion is the Steinhaus variant of the fourth-moment identity in [6, equation (2.10) and Remark 2.8], with coefficients $\sqrt{p_i} z_i$. We include the expansion to keep track of the order of the noncommuting factors. The two surviving pairings give

$$\mathbb{E}[(z^* z)^2] = A^2 + \sum_i p_i z_i^* B z_i - \sum_i p_i^2 (z_i^* z_i)^2. \quad (4.10)$$

The last sum subtracts the overlap of the pairings, where all four indices agree. Its summands are positive, so Equation (4.9) implies

$$\mathbb{E} \phi((z^* z)^2) \leq C. \quad (4.11)$$

We next use the functional-calculus truncation underlying [6, Lemma 2.6, equations (2.15)–(2.20)]. For $K > 0$, write $z = u|z|$ in its polar decomposition and put $z^{[K]} = u \min(|z|, K)$. Then $\|z^{[K]}\| \leq K$ and

$$\begin{aligned} \mathbb{E} \|(z - z^{[K]})\xi\|^2 &= \mathbb{E} \phi((|z| - K)_+^2) \\ &\leq K^{-2} \mathbb{E} \phi(|z|^4) \leq CK^{-2}. \end{aligned} \quad (4.12)$$

This controls the vector error on the side required for the bounded operator T . No estimate for the adjoint truncation error is needed.

Restoring the spectral band and selecting realizations. Define the random element $v = \mathcal{F}_{d_n}(z^{[K]}) \in M$. By Equation (4.3), every realization satisfies

$$\|v\| \leq C_1 K, \quad v\xi = f_{d_n}(D) z^{[K]} \xi, \quad (4.13)$$

and $v\xi$ is supported in $[s - 2d_n, s + 2d_n]$. Since $f_{d_n}(D)$ fixes $z\xi$ and has norm at most one, Equation (4.12) gives

$$\mathbb{E} \|v\xi - z\xi\|^2 \leq CK^{-2}.$$

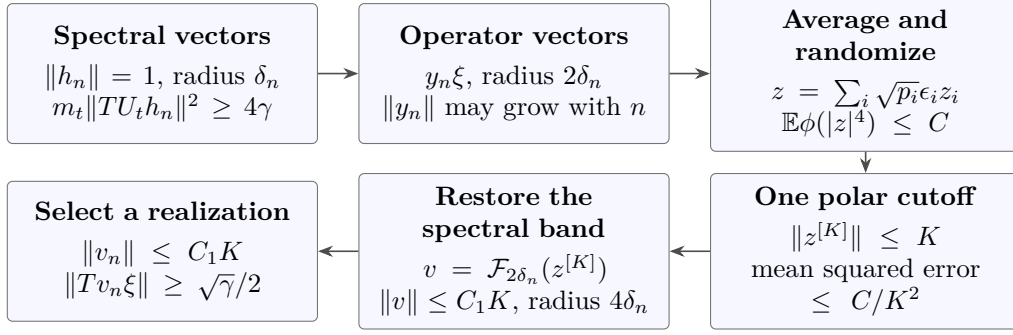


Figure 1: The construction in [Theorem 4.1](#), after passing to the subsequence with a positive averaged squared norm. Every band is centered at the fixed modular energy s . The polar cutoff need not preserve that band; the next Fourier filter restores it with radius $4\delta_n$. The constants C , K , and C_1 are independent of n . The cutoff error is averaged over the circle phases.

The triangle inequality in the Hilbert space of square-integrable random H -vectors now yields

$$(\mathbb{E}\|Tv\xi\|^2)^{1/2} \geq \sqrt{\gamma} - \frac{\|T\|\sqrt{C}}{K}. \quad (4.14)$$

Fix once and for all

$$K \geq 2\|T\|\sqrt{C/\gamma}.$$

The hypothesis implies $T \neq 0$, so K can be chosen positive. This choice is independent of n and ensures that the left-hand side of [Equation \(4.14\)](#) is at least $\sqrt{\gamma}/2$.

To select a realization, note that z depends norm-continuously on the finite phase torus and $z^{[K]} = zq_K(z^*z)$, where

$$q_K(r) = \min\{1, K/\sqrt{r}\} \quad (r > 0), \quad q_K(0) = 1.$$

Since q_K is continuous on $[0, \infty)$, continuous functional calculus makes the cutoff norm-continuous. The filter $\mathcal{F}_{d_n}$ is bounded and linear, so $\|Tv\xi\|^2$ is continuous on the compact phase torus. It therefore attains a maximum at least its expectation. Choose such a realization v_n for each n . Then

$$\|v_n\| \leq C_1 K, \quad \|Tv_n\xi\| \geq \frac{\sqrt{\gamma}}{2},$$

and its spectral support lies in $[s-2d_n, s+2d_n] = [s-4\delta_n, s+4\delta_n]$. Restoring the subsequence notation proves [Theorem 4.1](#) with $C_* = C_1 K$ and $\eta = \sqrt{\gamma}/2$. $\square$

[Figure 1](#) summarizes the construction. The averaging time depends on the initial representative, whereas the moment bound, cutoff threshold, and Fourier-kernel norm are independent of the width of its spectral band.

5 Spectral rigidity

We now prove [Theorem 1.2](#). Throughout this Section, assume its hypotheses and retain the modular notation and the symmetric mean from [Theorem 3.4](#). In particular, $M_\phi = \mathbb{C}1$. We use no ergodicity assumption on b .

5.1 From bounded recovery to an interval estimate

When $a\xi = \xi a^\flat$, the identity $wa\xi = R(a^\flat)w\xi$ for $w \in M$ writes the intertwining defect, at fixed s , as a bounded operator applied to $w\xi$.

Lemma 5.1 (Estimate at a spectral point). *Let $a, a^\flat \in M$ satisfy $a\xi = \xi a^\flat$. For $s \in \mathbb{R}$, set*

$$T_s = R(a^\flat) - L(b_s(a)) \in \mathcal{B}(H).$$

If $h_n \in H$ are unit vectors with D -spectral supports contained in $[s - d_n, s + d_n]$, where $d_n > 0$ and $d_n \rightarrow 0$, then

$$\|t \mapsto T_s U_t h_n\|_{\text{av}} \longrightarrow 0. \quad (5.1)$$

Proof. If the conclusion fails, [Theorem 4.1](#), applied to the fixed operator T_s , gives a uniformly bounded sequence $v_n \in M$ such that the D -spectral supports of $v_n \xi$ shrink to s and

$$\|T_s v_n \xi\| \geq \eta > 0 \quad \text{for every } n.$$

By [Theorem 3.1](#),

$$\|\phi v_n - e^s v_n \phi\|_{M_*} \longrightarrow 0.$$

The intertwining hypothesis (1.1) therefore gives $(v_n a - b_s(a) v_n) \xi \rightarrow 0$. Since left and right multiplication commute,

$$T_s v_n \xi = v_n \xi a^\flat - b_s(a) v_n \xi = (v_n a - b_s(a) v_n) \xi,$$

a contradiction. $\square$

The point estimate supplies no uniform rate in s . To pass to an interval, we use the commutation of the averaged quadratic form with the spectral projections of D .

Lemma 5.2 (Estimate on an interval). *Let $a, a^\flat \in M$ satisfy $a\xi = \xi a^\flat$. For every nonempty compact interval $I \subset \mathbb{R}$, every $c \in \mathbb{C}$, and every $h \in E_D(I)H$,*

$$\|t \mapsto (R(a^\flat) - cL(a))U_t h\|_{\text{av}} \leq \|h\| \sup_{s \in I} \|(b_s(a) - ca)\xi\|. \quad (5.2)$$

Proof. Put

$$S = R(a^\flat) - cL(a), \quad C = P(S^* S), \quad B = \sup_{s \in I} \|(b_s(a) - ca)\xi\|.$$

The number B is finite, and [Theorem 3.4](#) gives

$$\|t \mapsto S U_t h\|_{\text{av}}^2 = \langle Ch, h \rangle.$$

Suppose that the asserted estimate fails. By homogeneity there is a unit vector $h \in E_D(I)H$ with

$$\langle Ch, h \rangle = B^2 + \varepsilon \quad (\varepsilon > 0).$$

For each positive integer n , partition I into finitely many disjoint Borel intervals J of diameter at most $1/n$. Since C commutes with the spectral projections of D ,

$$\langle Ch, h \rangle = \sum_J \langle C E_D(J) h, E_D(J) h \rangle, \quad \sum_J \|E_D(J) h\|^2 = 1.$$

Consequently, one nonzero spectral piece, after normalization, gives a unit vector h_n supported in $\overline{J_n}$ such that

$$\langle Ch_n, h_n \rangle \geq B^2 + \varepsilon. \quad (5.3)$$

By compactness of I , after passing to a subsequence these supports lie in intervals shrinking to a single point $s \in I$. This also covers a singleton interval I .

Write $z = b_s(a) - ca$, so that $S = T_s + L(z)$. By [Theorem 5.1](#), $\|t \mapsto T_s U_t h_n\|_{\text{av}} \rightarrow 0$. On the other hand, [Theorem 3.4](#) gives the exact identity

$$\|t \mapsto L(z) U_t h_n\|_{\text{av}}^2 = \langle P(L(z^* z)) h_n, h_n \rangle = \phi(z^* z) \leq B^2.$$

The triangle inequality now yields $\limsup_n \|t \mapsto S U_t h_n\|_{\text{av}} \leq B$, contrary to [\(5.3\)](#). $\square$

5.2 The joint spectral pairing

State preservation gives a unitary implementation

$$V_s(a\xi) = b_s(a)\xi \quad (a \in M).$$

It is strongly continuous by the u -continuity of b : for $a \in M$,

$$\|V_s(a\xi) - a\xi\|^2 = 2\phi(a^* a) - 2\operatorname{Re} \phi(a^* b_s(a)) \rightarrow 0 \quad (s \rightarrow 0),$$

and density extends this continuity to H . Since b_s preserves ϕ , it commutes with the modular automorphism group. Thus $V_s U_t = U_t V_s$ for all s, t . Write $V_s = e^{isQ}$. The self-adjoint operators D and Q strongly commute; let E be their joint spectral resolution and let $\mathcal{S} \subset \mathbb{R}^2$ be its closed support.

The joint spectral measure is concentrated on $\mathcal{S}$. Indeed, its complement is a union of open sets with zero spectral projection, and the countable base of $\mathbb{R}^2$ reduces this to a countable union. Hence $E(\mathcal{S}) = 1$, without any separability assumption on H .

Lemma 5.3 (Joint spectral pairing). *For every $(r, p), (q, \ell) \in \mathcal{S}$,*

$$\exp(i(r\ell + qp)) = 1. \tag{5.4}$$

Proof. Fix $\delta > 0$ and write $I_v = [v - \delta, v + \delta]$. We first choose nonzero $x, a \in M$ such that

$$x\xi \in E(I_r \times I_p)H, \quad a\xi \in E(I_q \times I_\ell)H.$$

Here is the justification for this bounded localization. Choose nonnegative smooth functions f, g supported in the interiors of I_r, I_p and equal to one on smaller neighborhoods of r, p . The spectral projection of the resulting smaller rectangle is nonzero by the definition of $\mathcal{S}$. Consequently, $f(D)g(Q) \neq 0$. Density of $M\xi$ supplies $y \in M$ with $f(D)g(Q)y\xi \neq 0$. Fourier integration along b realizes $g(Q)y\xi$ as a vector in $M\xi$: if $g(u) = \int_{\mathbb{R}} \widehat{g}(t)e^{itu} dt$, use the bounded weak integral $\int_{\mathbb{R}} \widehat{g}(t)b_t(y) dt$. The subsequent filter $f(D)$ also preserves $M\xi$ by [Theorem 3.2](#). Their composition therefore gives the required x . The construction of a is identical. No bound uniform in δ on the resulting operators is needed.

Both vectors have compact modular spectral support. By [Theorem 3.2](#), choose $x^b, a^b \in M$ with

$$x\xi = \xi x^b, \quad a\xi = \xi a^b.$$

Set

$$c = e^{ir\ell}, \quad d = e^{iqp}, \quad H_0 = \|a\xi\| \|x\xi\| > 0,$$

and consider the bounded continuous vector functions

$$F(t) = \sigma_t(x)a\xi, \quad G(t) = a\sigma_t(x)\xi.$$

Since

$$F(t) - cG(t) = (R(a^b) - cL(a))U_t(x\xi),$$

[Theorem 5.2](#) and the Q -spectral support of $a\xi$ give

$$\begin{aligned} \|F - cG\|_{\text{av}} &\leq \|x\xi\| \sup_{s \in I_r} \|(b_s(a) - ca)\xi\| \\ &\leq H_0 \sup_{\substack{s \in I_r \\ u \in I_\ell}} |e^{isu} - c|. \end{aligned} \quad (5.5)$$

Interchanging a and x in the same argument yields

$$\|t \mapsto \sigma_t(a)x\xi - dx\sigma_t(a)\xi\|_{\text{av}} \leq H_0 \sup_{\substack{s \in I_q \\ u \in I_p}} |e^{isu} - d|.$$

The pointwise identity

$$U_t(\sigma_{-t}(a)x\xi - dx\sigma_{-t}(a)\xi) = G(t) - dF(t) \quad (5.6)$$

and reflection invariance of the mean identify the seminorm on the left with $\|G - dF\|_{\text{av}}$. In particular, the phase is still d : the unitary U_t is complex linear. We have proved

$$\|G - dF\|_{\text{av}} \leq H_0 \sup_{\substack{s \in I_q \\ u \in I_p}} |e^{isu} - d|. \quad (5.7)$$

The averaged norm of F has the same factor H_0 as the two error bounds. By reflection invariance and [Theorem 3.4](#),

$$\begin{aligned} \|F\|_{\text{av}}^2 &= m_t \langle L(\sigma_t(x^*x))a\xi, a\xi \rangle \\ &= \langle P(L(x^*x))a\xi, a\xi \rangle = \phi(x^*x)\phi(a^*a) = H_0^2. \end{aligned} \quad (5.8)$$

Using $|c| = 1$ and

$$(1 - cd)F = (F - cG) + c(G - dF),$$

combine (5.5) and (5.7), and divide by H_0 , to obtain

$$|1 - cd| \leq \sup_{\substack{s \in I_r \\ u \in I_\ell}} |e^{isu} - e^{ir\ell}| + \sup_{\substack{s \in I_q \\ u \in I_p}} |e^{isu} - e^{iqp}|.$$

The right side tends to zero as $\delta \rightarrow 0$. For example, the first supremum is at most $\delta(|r| + |\ell|) + \delta^2$, by $|e^{iv} - e^{iw}| \leq |v - w|$ for real v, w . The second has the analogous bound. Since c and d depend only on the original support points, this proves (5.4). $\square$

5.3 Conclusion of the proof

We first record the restriction that a scalar modular centralizer imposes on modular eigenvalues. This is the weak-mixing argument of Marrakchi and Vaes [14, Lemma 2.1]; we include the needed conclusion and its proof in our spectral notation. The bounded operator extraction is useful because a spectral projection on H need not itself preserve $M\xi$.

Lemma 5.4 (Modular eigenvalues). *If $M_\phi = \mathbb{C}1$, then D has no nonzero eigenvalue. If, in addition, D has pure point spectrum, then $M = \mathbb{C}1$.*

Proof. Suppose $E_D(\{r\}) \neq 0$. Choose $x \in M$ such that $\eta = E_D(\{r\})x\xi \neq 0$, using density of $M\xi$. Let

$$y_k = \int_{\mathbb{R}} e^{-irt} \sigma_t(x) d\mu_k(t),$$

where μ_k is uniform probability measure on $[-k, k]$. These bounded weak integrals satisfy $\|y_k\| \leq \|x\|$. The Hilbert-space mean ergodic theorem for $t \mapsto e^{-irt}U_t$ gives

$$y_k\xi \longrightarrow \eta \quad \text{in } H.$$

Take a weak operator convergent subnet of (y_k) , with limit $y \in M$. Its vector limit is $y\xi = \eta \neq 0$. Therefore, for every $t \in \mathbb{R}$,

$$\sigma_t(y)\xi = U_t y\xi = e^{irt}y\xi.$$

Since ξ is separating, $\sigma_t(y) = e^{irt}y$.

It follows that $y^*y, yy^* \in M_\phi = \mathbb{C}1$. Their scalar values coincide because $\|y^*y\| = \|yy^*\|$, and are strictly positive. Thus y is a nonzero scalar multiple of a unitary, and $\phi(y^*y) = \phi(yy^*) > 0$. The modular identity (3.1) now gives

$$\phi(yy^*) = \|\xi y\|^2 = \|e^{D/2}y\xi\|^2 = e^r\|y\xi\|^2 = e^r\phi(y^*y).$$

Hence $r = 0$. If D has pure point spectrum, all of H is consequently its zero eigenspace, so $D = 0$. Then the modular action is trivial and $M = M_\phi = \mathbb{C}1$. $\square$

Proof of Theorem 1.2. By Theorem 5.3, the symmetric bilinear form

$$\mathfrak{b}((r, p), (q, \ell)) = r\ell + qp$$

takes values in $2\pi\mathbb{Z}$ on every pair of points of $\mathcal{S}$. We claim that $\mathcal{S}$ is countable or is contained in a coordinate axis.

If $\mathcal{S}$ contains linearly independent vectors v_1, v_2 , the map

$$z \longmapsto (\mathfrak{b}(v_1, z), \mathfrak{b}(v_2, z))$$

is an invertible real linear map on $\mathbb{R}^2$, since the form is nondegenerate and v_1, v_2 are independent. It sends $\mathcal{S}$ into the countable set $(2\pi\mathbb{Z})^2$, proving countability. Otherwise $\mathcal{S}$ is contained in a line through the origin. If this line is not a coordinate axis, write its points as $t(\alpha, \beta)$ with $\alpha\beta \neq 0$. Pairing each point with itself gives

$$\alpha\beta t^2 \in \pi\mathbb{Z}.$$

There are at most two real values of t for each integer, again proving countability. The zero-dimensional case is contained in both axes. This proves the claim.

If $\mathcal{S}$ is countable, its spectral projections at singletons sum strongly to 1, because $E(\mathcal{S}) = 1$. Each such projection has range in an eigenspace of D , so D has pure point spectrum. By Theorem 5.4, $M = \mathbb{C}1$, and b is trivial.

If instead $\mathcal{S} \subset \{0\} \times \mathbb{R}$, then $D = 0$, again giving $M = M_\phi = \mathbb{C}1$ and triviality of b . In the remaining case $\mathcal{S} \subset \mathbb{R} \times \{0\}$, we have $Q = 0$. Thus $V_s = 1$ for every s , and

$$b_s(a)\xi = V_s(a\xi) = a\xi \quad (a \in M).$$

Separation by ξ gives $b_s(a) = a$. In all cases b is the trivial action, as asserted. $\square$

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